EV Adoption & Charging Infrastructure Analysis

- Introduction

Electric vehicles (EVs) are becoming increasingly popular due to rising environmental concerns and government initiatives to reduce carbon emissions. As EV adoption grows, the need for a strong charging infrastructure becomes essential to support this transition.

This project analyzes EV adoption trends and charging station availability across major cities in India from 2020 to 2025. The objective is to identify growth patterns, regional adoption differences, and infrastructure gaps using data analytics tools.

The analysis was performed using Microsoft Excel for data preprocessing and Power BI for visualization and dashboard development.

- Problem Statement

With the rapid increase in electric vehicle adoption, governments and energy providers need to ensure that charging infrastructure grows at a similar pace. However, some regions may have high EV adoption but insufficient charging stations, creating infrastructure challenges.

This project aims to answer the following questions:

• How has EV adoption changed over time?
• Which cities are leading in EV adoption?
• Is the charging infrastructure keeping up with EV growth?
• Which regions require additional infrastructure investment?

- Objectives of the Project

The main objectives of this project are:

- Analyze electric vehicle adoption trends from 2020 to 2025.
- Identify cities with the highest EV registrations.
- Compare charging infrastructure availability across regions.
- Detect infrastructure gaps using EV-to-charging-station ratios.
- Provide insights that can support policy and infrastructure planning.

- Dataset Description

The dataset used in this project contains EV registration and charging station information for major Indian cities. The dataset includes the following variables:

Column	Description
Year	Year of EV registration
Country	Country name
State	State where EV is registered
City	City name
EV Type	Type of EV (Battery EV or Hybrid)
Vehicle Category	Category of vehicle
Charging Stations	Number of EV charging stations
EV Registrations	Total EV registrations
Population	Population of the region

- Data Cleaning and Preparation

Before analysis, the dataset was cleaned and transformed to ensure data accuracy.

The following steps were performed:

• Removed duplicate entries from the dataset.
• Checked for missing values and handled them appropriately.
• Standardized EV type categories such as “Battery EV” and “Hybrid”.
• Verified numeric fields like EV registrations and charging stations.
• Created calculated fields such as **EV Adoption Rate** and **Charging Ratio**.

Example calculated columns:

- EV Adoption Rate = EV Registrations / Population
- Charging Ratio = Charging Stations / EV Registrations
- These metrics help evaluate EV penetration and infrastructure availability.

- Tools Used

Tool	Purpose
Microsoft Excel	Data cleaning and preprocessing
Power BI	Data visualization and dashboard creation
GitHub	Project documentation and portfolio

- Dashboard Overview

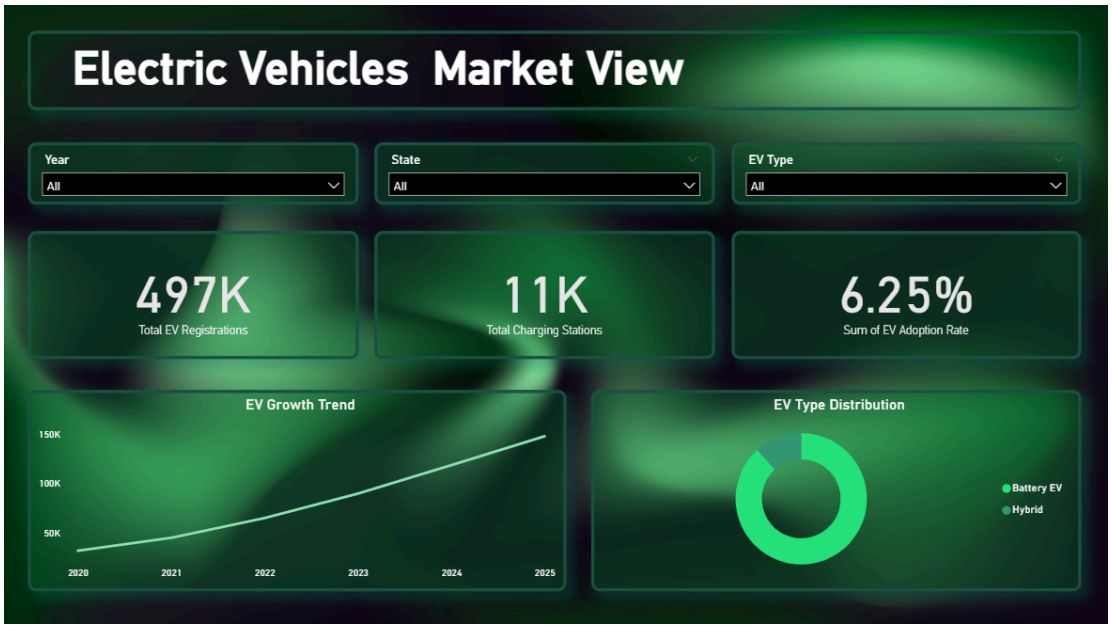
1: EV Market Overview

This page provides a high-level overview of the EV market.

Visualizations include:

• KPI cards showing total EV registrations, total charging stations, EV adoption rate, and average charging ratio.
• A line chart displaying EV adoption trends from 2020 to 2025.
• A donut chart illustrating the distribution of EV types.

This page helps understand overall market growth and EV type popularity.



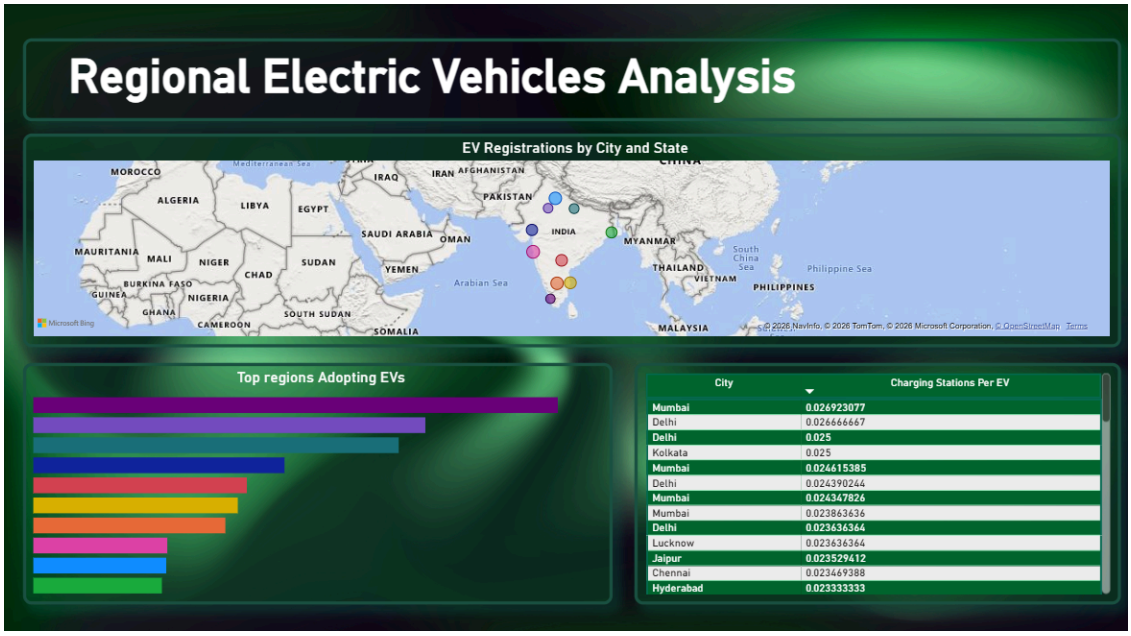
2: Regional EV Analysis

This page focuses on geographic analysis.

Visualizations include:

- A map showing EV registrations by city.
- A bar chart highlighting cities with the highest EV adoption.
- A table comparing EV registrations and charging stations across regions.

This page identifies areas where EV adoption is growing rapidly.



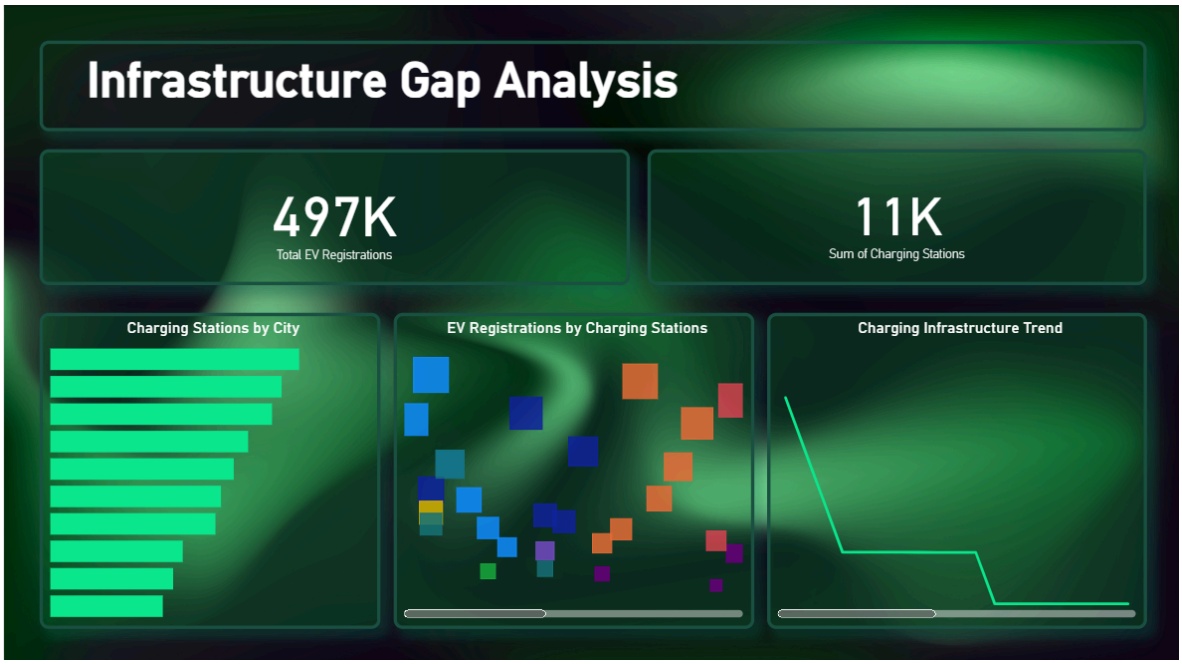
3: Infrastructure Gap Analysis

This page highlights regions where EV adoption may be outpacing infrastructure.

Visualizations include:

- A scatter plot comparing EV registrations with charging stations.
- A bar chart showing cities with the lowest charging ratio.

These visuals help detect infrastructure gaps and support planning decisions.



- <u>Key Insights</u>

1. EV adoption has grown significantly between 2020 and 2025.
2. Major metropolitan cities such as Delhi, Mumbai, and Bengaluru lead in EV registrations.
3. Some emerging cities show high EV adoption but limited charging infrastructure.
4. Battery electric vehicles dominate the EV market compared to hybrid vehicles.

- <u>Recommendations</u>

Based on the analysis, the following recommendations can be made:

• Expand charging infrastructure in cities with low charging ratios.
• Encourage EV adoption in smaller cities through incentives and awareness programs.
• Invest in public charging networks to support growing EV demand.
• Monitor infrastructure development to ensure it keeps pace with EV adoption.

- <u>Conclusion</u>

The analysis shows that electric vehicle adoption is increasing rapidly, especially in major urban regions. While charging infrastructure has improved over time, certain cities still face infrastructure gaps. Addressing these gaps will be essential to support future EV growth and ensure a smooth transition toward sustainable transportation.

This project demonstrates how data analytics tools like Excel and Power BI can be used to analyze trends, identify infrastructure needs, and generate actionable insights.